WORKED SOLUTION

STEP III Specimen paper, Question 3

Pure • Complex Triangle Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q3

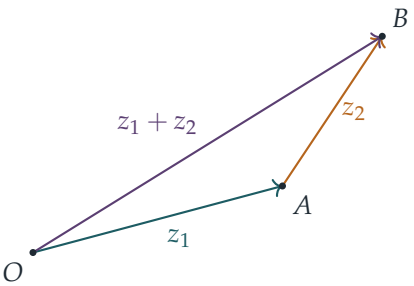
The key idea

The inequality is the triangle inequality: no side of a triangle is longer than the other two put together. Applied repeatedly, it lets us bound the modulus of a whole sum by the sum of the moduli. If the left-hand side of the equation is always strictly smaller than 1 in modulus when $|z| \leq \frac{1}{3}$, it can never equal 1, and a geometric series does the bounding.

Solution

The geometric meaning

In the Argand diagram, think of z_1 as the displacement from the origin O to a point A, and z_2 as the displacement from A to a point B. Then $z_1 + z_2$ is the displacement from O straight to B.



The lengths of the sides of triangle OAB are $|z_1|$, $|z_2|$ and $|z_1 + z_2|$. The inequality says that going straight from O to B is never longer than going via A: one side of a triangle is at most the sum of the other two. Equality holds exactly when the triangle is flat with A lying on the segment OB, that is, when z_1 and z_2 point in the same direction (one is a non-negative real multiple of the other, or one of them is zero).

By induction, the same holds for any finite number of terms:

$$|w_1 + w_2 + \cdots + w_n| \leq |w_1| + |w_2| + \cdots + |w_n|,$$

since $|w_1 + \cdots + w_{k+1}| \leq |w_1 + \cdots + w_k| + |w_{k+1}|$. Geometrically, a path made of n straight steps is at least as long as the straight line from its start to its end.

No solutions with $|z| \leq \frac{1}{3}$

Suppose $|z| \leq \frac{1}{3}$ and $|a_i| \leq 2$ for every i. Using the extended triangle inequality, then $|ab| = |a||b|$,

$$\begin{aligned}\left|a_1 z + a_2 z^2 + \cdots + a_n z^n\right| &\leq |a_1||z| + |a_2||z|^2 + \cdots + |a_n||z|^n \\ &\leq 2\left(\tfrac{1}{3} + \tfrac{1}{3^2} + \cdots + \tfrac{1}{3^n}\right),\end{aligned}$$

where the second step uses $|a_i| \leq 2$ and $|z|^k \leq \left(\frac{1}{3}\right)^k$ for each k (all the quantities involved are non-negative, so we may bound term by term).

The bracket is a finite geometric series with first term $\frac{1}{3}$ and ratio $\frac{1}{3}$:

$$2\sum_{k=1}^{n} \frac{1}{3^k} = 2 \cdot \frac{\frac{1}{3}\left(1 - 3^{-n}\right)}{1 - \frac{1}{3}} = 1 - \frac{1}{3^n} < 1.$$

Ask yourself Why is the inequality strict, when each step above used only $\leq$?

The strictness comes from the series, not from the triangle inequality. Because the sum stops at n terms, it falls short of its infinite total 1 by exactly $3^{-n} > 0$. So whatever the a_i and whatever z with $|z| \leq \frac{1}{3}$,

$$\left|a_1 z + a_2 z^2 + \cdots + a_n z^n\right| \leq 1 - 3^{-n} < 1.$$

But if z were a solution, the left-hand side would be $|1| = 1$. This is a contradiction, so the equation has no solutions with $|z| \leq \frac{1}{3}$, as required.

Checks

- The worst case is $a_1 = \cdots = a_n = 2$ and $z = \frac{1}{3}$, when every inequality becomes an equality and the left-hand side is $1 - 3^{-n}$. For $n = 10$ this is $0.99998\ldots$, just short of 1.
- The constant $\frac{1}{3}$ cannot be improved. With every $a_i = 2$, the real function $2(x + x^2 + \cdots + x^n) - 1$ is negative at $x = \frac{1}{3}$, and for large n it is positive just beyond $\frac{1}{3}$, so there are real roots as close to $\frac{1}{3}$ as we like.
- A numerical test of 200 000 random choices of $n \leq 30$, of a_i with $|a_i| \leq 2$ and of z with $|z| \leq \frac{1}{3}$ never gave a left-hand side of modulus 1 or more.

Common slips

- Bounding by the infinite series and getting ≤ 1, which does not rule out equality. You need the strict inequality, and it comes from the series being finite.
- Writing $|a_1 z + \cdots + a_n z^n| \leq |a_1 z| + \cdots$ without saying why: this is the triangle inequality extended to n terms, which is the reason the first part is there.

Beyond STEP

This is the simplest example of a *root-free region*: bounds on the coefficients of a polynomial keep its roots away from a disc. The same estimate, run the other way, gives Cauchy's bound that every root of $z^n + c_{n-1}z^{n-1} + \cdots + c_0$ has modulus less than $1 + \max |c_k|$, and in complex analysis Rouché's theorem turns arguments like this into exact counts of roots inside a circle. For a beautifully geometric account, read Needham, *Visual Complex Analysis*.

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